

Algebra General Exam - August 2025

Your UVa ID Number:

- There is a total of 80 points in 8 problems.
- Please provide *complete proofs* and justify *every* statement that you make.
- Make sure that the solution to every problem is a *continuous text* written legibly and in the correct order.
- If you refer to a *standard theorem*, please, *state* this theorem clearly and fully, and *justify* that this theorem applies in the situation at hand.
- Vague statements and hand-waving arguments will not be appreciated.
- You may assume the validity of one part of the problem to do another part even if you unable to prove the first part (in which case you, of course, will not get credit for it).

DO EACH PROBLEM ON A SEPARATE SHEET OF PAPER, AND STAPLE THEM TOGETHER IN THE CORRECT ORDER BEFORE TURNING THE EXAM IN.

Sign below the pledge:

“On my honor, I pledge that I have neither given nor received help on this assignment.”

Signature (Use your ID number instead):

Problem 1

Let $\mathbb{F}_q$ be a finite field with q elements.

(a) Find the orders of the general linear group $\mathrm{GL}_2(\mathbb{F}_q)$ and special linear group $\mathrm{SL}_2(\mathbb{F}_q)$: (4 points)

$$\mathrm{GL}_2(\mathbb{F}_q) := \{A \in \mathrm{M}_2(\mathbb{F}_q) \mid \det A \neq 0\}, \quad \mathrm{SL}_2(\mathbb{F}_q) := \{A \in \mathrm{M}_2(\mathbb{F}_q) \mid \det A = 1\}.$$

(b) Show that $\mathrm{SL}_2(\mathbb{F}_q)$ contains elements of orders $q - 1$ and $q + 1$. (*Hint.* To find an element of order $q + 1$, use the extension $\mathbb{F}_{q^2} / \mathbb{F}_q$.) (6 points)

Problem 2

As usual, S_6 and A_6 will denote the symmetric and alternating groups on 6 elements.

(a) Let $\sigma = (123)(456) \in A_6$. Find the sizes of the conjugacy classes $C_{S_6}(\sigma)$ and $C_{A_6}(\sigma)$. (5 points)

(b) Let $\tau = (12345) \in A_6$. Find the sizes of the conjugacy classes $C_{S_6}(\tau)$ and $C_{A_6}(\tau)$. (5 points)

Problem 3

The symmetric group $G = S_4$ has the following character table, where for each conjugacy class C_i we indicate a representative and its size c_i .

Character table of the symmetric group S_4

C_i	(1)	(12)	(12)(34)	(123)	(1234)
c_i	1	6	3	8	6
χ_1	1	1	1	1	1
χ_2	1	-1	1	1	-1
χ_3	2	0	2	-1	0
χ_4	3	1	-1	0	-1
χ_5	3	-1	-1	0	1

(a) Use the table to determine the decomposition of the representation $\rho_4 \otimes \rho_5$ into the direct sum of irreducible representations, where ρ_i denotes the irreducible representation of G with character χ_i . (5 points)

(b) Let $G = S_n$ with $n \geq 4$. Let ρ be the standard $(n - 1)$ -dimensional representation of G ("permutation representation minus the trivial representation"), and let σ be the sign representation (in other words, the 1-dimensional representation whose kernel is the alternating group A_n). Show that the representations ρ and $\rho \otimes \sigma$ are not equivalent. (5 points)

Problem 4

Let $M = \mathbb{Z}^n$ be a free $\mathbb{Z}$ -module of rank $n \geq 2$. We recall that the group

$\text{SL}_n(\mathbb{Z}) = \{A \in \text{M}_n(\mathbb{Z}) \mid \det A = 1\}$ acts on M by automorphisms through matrix multiplication. Set

$$X = \{(a_1, \dots, a_n) \in M \setminus \{0\} \mid \text{g.c.d}(a_1, \dots, a_n) = 1\}.$$

(a) Show that for any $w \in X$ there exists a basis $v_1, \dots, v_n$ of M such that $v_1 = w$. (4 points)

(b) Let $w = (6, 10, 15)$. Find a basis v_1, v_2, v_3 of $M = \mathbb{Z}^3$ such that $v_1 = w$. (3 points)

(c) Use part (a) to show that $\text{SL}_n(\mathbb{Z})$ acts on X transitively. (3 points)

Problem 5

Let F be a field of characteristic 5. Classify up to conjugacy all (5×5) -matrices X over F satisfying $X^5 = I_5$. (10 points)

Problem 6

Let $R = \mathbb{C}[x, y]$ be the ring of complex polynomials in two variables. For an element $a \in R$, we let (a) denote the principal ideal generated by a . Prove the following properties of the ring R .

(a) While the g.c.d. of x and y is 1, show that the natural homomorphism

$$R \rightarrow R/(x) \times R/(y)$$

is not surjective. (4 points)

(b) Prove that no principal ideal of R is maximal. (6 points)

Problem 7

Let F be a field, and let $f(x), g(x) \in F[x]$ be nonconstant polynomials. Then the quotients $F[x]/(f(x))$ and $F[x]/(g(x))$ can be considered as $F[x]$ -modules/algebras as well as F -vector spaces/algebras.

(a) Use the universal property of tensor product to show that there exists a surjective homomorphism of F -algebras (4 points):

$$\gamma : \frac{F[x]}{(f(x))} \otimes_F \frac{F[x]}{(g(x))} \rightarrow \frac{F[x]}{(f(x))} \otimes_{F[x]} \frac{F[x]}{(g(x))}.$$

(b) Show that γ is an isomorphism if and only if f and g are proportional linear polynomials. (6 points)

Problem 8

Let E/K be a cyclic Galois extension of degree n with Galois group $G = \langle \sigma \rangle$. Show that there exists $a \in E$ such that $a, \sigma(a), \dots, \sigma^{n-1}(a)$ is a K -basis of E . (*Hint.* Consider σ as a K -linear automorphism of the K -vector space E . Determine the minimal polynomial of σ , and use this information to conclude that E has a “cyclic” basis for σ .) (10 points)